Sahaj Bhadja

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Summary

Software engineer with 2+ years building reliable applications, scientific data workflows, REST services, and containerized systems using Python, Java, C++, PostgreSQL, Docker, object-oriented design, automated testing, and production debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant services on Azure SQL and Blob Storage for 500+ users, implementing normalized schemas, secure access controls, and indexed queries that reduced dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workers for 1,000+ monthly jobs with explicit states, retries, validation, and failure isolation, tripling throughput while strengthening debugging visibility and overall production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows processing 10,000+ daily requests, integrating secure PDF, DOCX, and ZIP ingestion with automated analysis while maintaining stable production application performance.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading and fault-tolerant execution, reducing turnaround by 60% while improving API reliability under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a reproducible scientific pipeline processing 100,000+ trajectory logs for autonomous-driving simulations, improving long-horizon planning by 18% through structured data preparation, model evaluation, and experiment tracking.
- Architected a spatial-temporal encoder combining HD-map geometry and historical sensor-derived trajectories through multi-head attention, improving closed-loop planning stability by 15% across diverse, repeatable simulated system operating scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release quality while validating functionality across Windows, Linux, and macOS environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN application with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex operational requirements into maintainable real-time software workflows.
- Automated Docker-based testing and staged deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, technical documentation, and defect resolution across frequent releases.

Skills

Languages & Design: Python, Java, C++, JavaScript, TypeScript, SQL, Bash, object-oriented programming, data structures, algorithms, design patterns, system design, concurrency, multithreading; **Backend & Data:** FastAPI, Flask, Node.js, REST APIs, WebSockets, microservices, PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, data modeling, ETL, data validation; **Cloud & Delivery:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, Linux; **Quality & Operations:** PyTest, Mockito, unit testing, integration testing, system testing, automated test suites, debugging, root-cause analysis, monitoring, performance optimization, fault tolerance, documentation, Agile/Scrum

Projects

HMIS Live | *MERN, TypeScript, REST APIs, Sentry, Twilio*
- Implemented consent-based healthcare workflows, secure API integrations, automated notifications, and Sentry monitoring, supporting 100+ daily appointments while reducing administrative workload by 30% across reliable clinical software operations.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker, Node.js*
- Built a database application supporting 10+ SQL and NoSQL systems, combining schema exploration, validated query workflows, and Docker-based cross-platform deployment across Windows, macOS, and Linux development environments.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ simulated process combinations while reducing algorithm-debugging time by 20%.